

Lecture/Week 8

Numerical Integration

Introduction

$$\int f(x) dx$$

Example

$$f(x) = x^2$$

$$\int f(x) dx = \int x^2 dx = \frac{x^3}{3} + C$$

$$f(x) = \sin x$$

$$\int f(x) dx = \int \sin x dx = -\cos x + C$$

* Integration of a given function is another function

** Numerically it is not possible.

When we integrate a function and answer is a number?

→ In the case of definite Integral
Integration with a limit

$$\int_a^b f(x) dx$$

Example $f(x) = x^2$

$$a = 0 \quad b = 2$$

$$\int_a^b f(x) dx = \int_0^2 x^2 dx = \left| \frac{x^3}{3} \right|_0^2 = \frac{(2)^3}{3} - 0$$

$$= \frac{8}{3} \text{ Ans.}$$

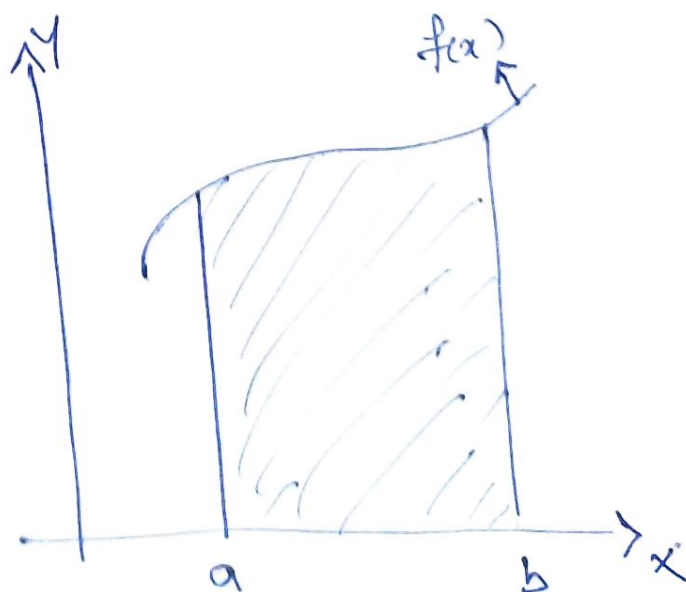
Numerical Integration is possible for definite integrals.

Methods

- 1) Rectangular Rule
- 2) Trapezoidal Rule
- 3) Simpson's Rule.

$$\int_a^b f(x) dx = \text{Area under the curve } f(x) \text{ from point 'a' to 'b'}$$

[~~shaded~~
Shaded Area]

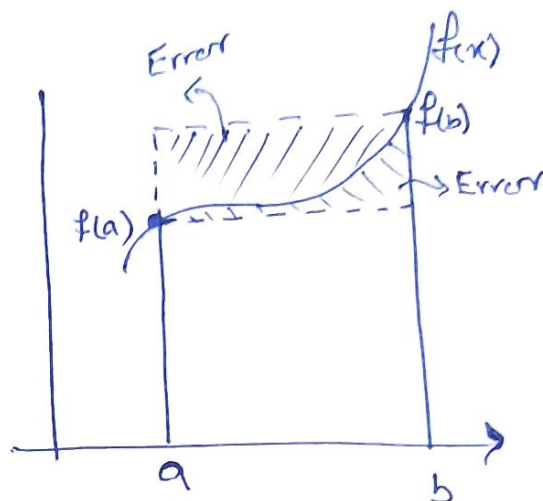


Task is to approximate the area.

$$\int_a^b f(x) dx \approx \text{Area under the curve}$$

Rectangular Rule

$$\begin{aligned} \int_a^b f(x) dx &\approx \text{Area of a rectangle} \\ &= (b-a) * f(a) \\ &\quad \text{or} \\ &= (b-a) * f(b) \end{aligned}$$



Example

$$\int_1^2 e^{-x^2/2} dx$$

Solution $f(x) = e^{-x^2/2}$

$$a=1$$

$$b=2$$

$$\int_1^2 e^{-x^2/2} dx = (b-a) * f(a) = (2-1) e^{-1/2} = 0.606531$$

$$\int_1^2 e^{-x^2/2} dx = (b-a) * f(b) = (2-1) e^{-2} = 0.135335$$

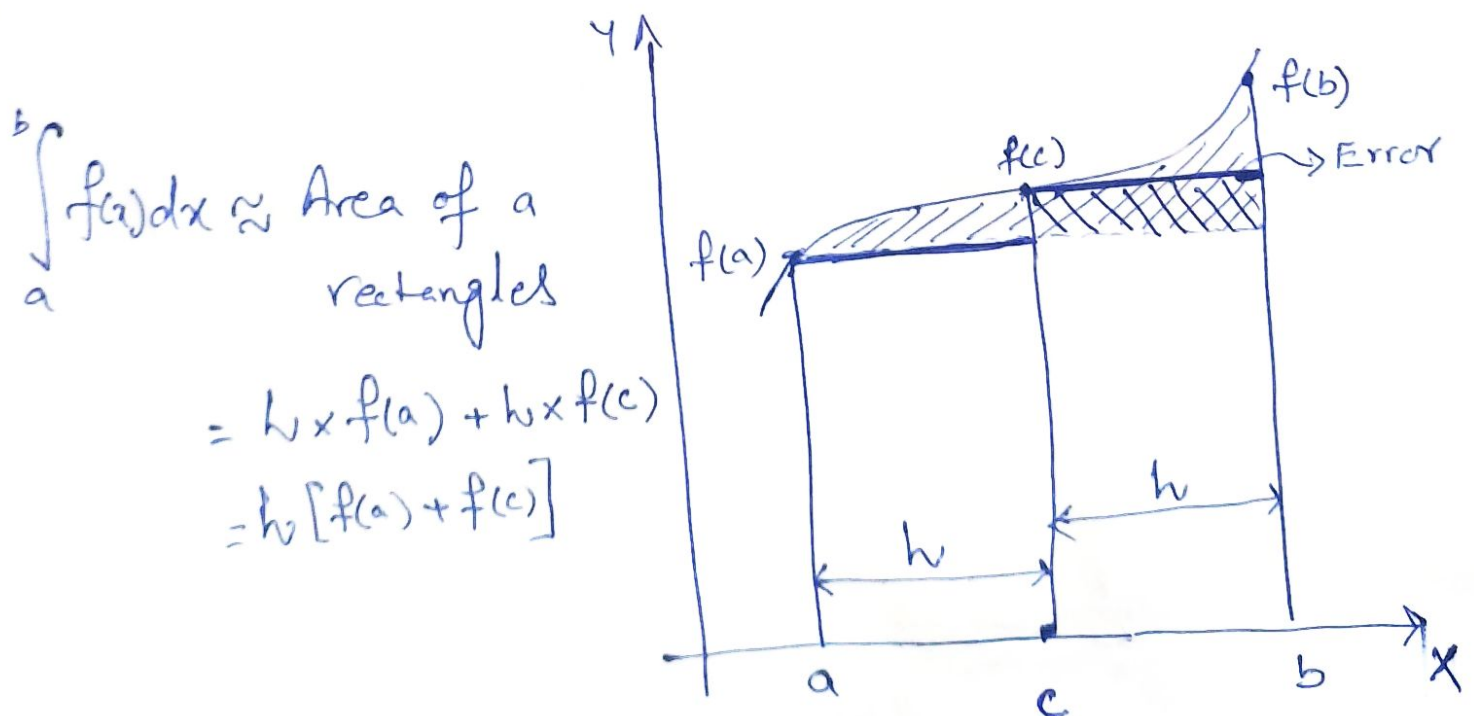
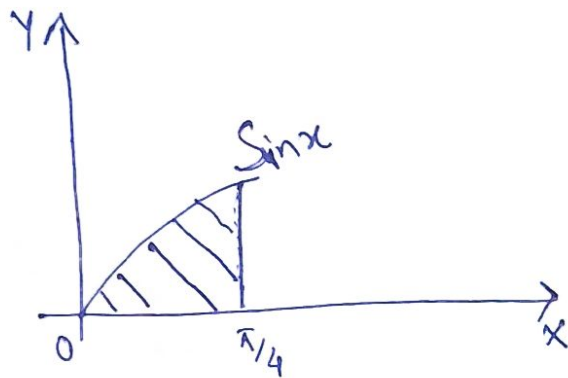
Example (2)

$$\int_0^{\pi/4} \sin x \, dx$$

$$f(x) = \sin x \quad a = 0 \quad b = \pi/4$$

$$\int_0^{\pi/4} \sin x \, dx = (b-a) \times f(a) = (\pi/4 - 0) \times \sin(0) = 0$$

$$\int_0^{\pi/4} \sin x \, dx = (b-a) \times f(b) = (\pi/4 - 0) \sin \pi/4 = \frac{\pi}{4} \cdot \frac{1}{\sqrt{2}} = \dots$$



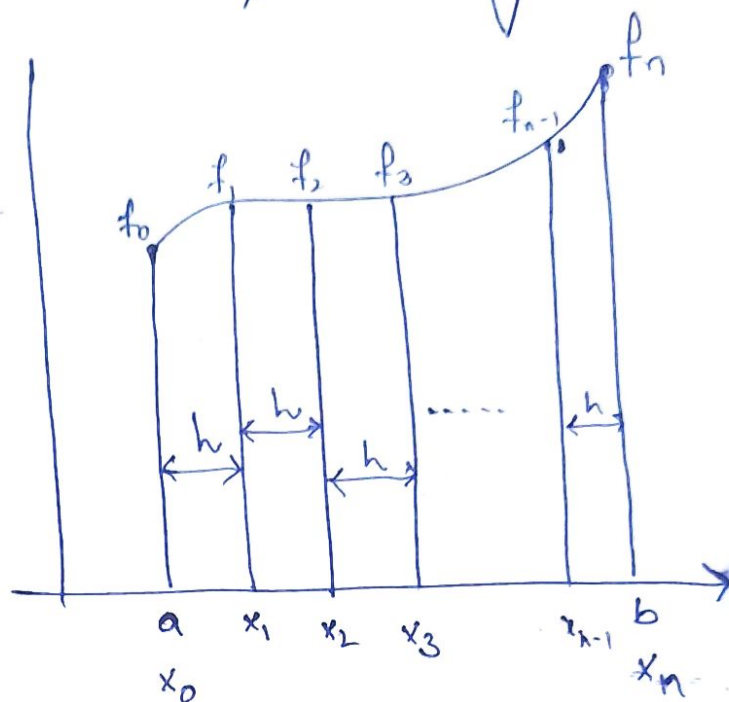
Dividing area into 'n' number of rectangles.

$$\int_a^b f(x) dx = h \times f_0 + h \times f_1 + h \times f_2 + \dots + h \times f_{n-1}$$

$$= h [f_0 + f_1 + f_2 + \dots + f_{n-1}]$$

General formular

or



where $f(x_0) = f_0$; $f(x_1) = f_1$
 $f(x_2) = f_2$ $f(x_n) = f_n$

$$h = \frac{b-a}{n}$$

$$= \frac{x_n - x_0}{n}$$

$$\int_a^b f(x) dx = h [f_1 + f_2 + \dots + f_n]$$

$$\int_a^b f(x) dx = h [f_0 + f_1 + f_2 + \dots + f_{n-1}]$$

Example Approximate $\int_0^2 \cosh x \, dx$ by using 4 subintervals in the rectangular rule.

Solution $f(x) = \cosh x$ $x_0 = 0$ $x_n = 2$ $n = 4$
 $h = \frac{x_n - x_0}{n} = \frac{2 - 0}{4} = 0.5$

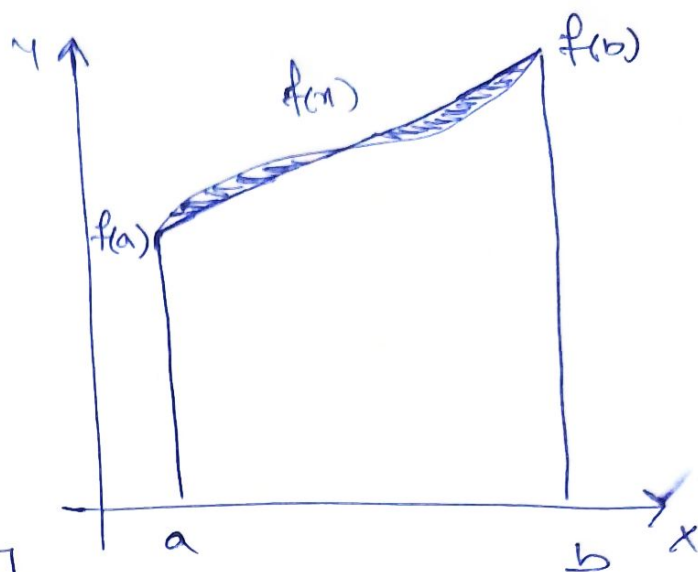
<u>X</u>	<u>f(x)</u>
$x_0 = 0$	$f_0 = f(x_0) = \cosh(0) = 1$
$x_1 = 0.5$	$f_1 = f(x_1) = \cosh(0.5) = 1.127626$
$x_2 = 1.0$	$f_2 = f(x_2) = \cosh(1.0) = 1.543081$
$x_3 = 1.5$	$f_3 = f(x_3) = \cosh(1.5) = 2.352410$
$x_4 = 2.0$	$f_4 = f(x_4) = \cosh(2) = 3.762196$

$$\begin{aligned}
 \int_0^2 \cosh x \, dx &= h [f_0 + f_1 + f_2 + f_3] \\
 &= 0.5 [1 + 1.127626 + 1.543081 + 2.352410] \\
 &= \underline{\underline{3.01156}} \\
 &\quad \text{Ans.}
 \end{aligned}$$

Trapezoidal Rule

⑦

$$\int_a^b f(x) dx \approx \text{Area of a trapezium}$$



$$\int_a^b f(x) dx \approx \frac{1}{2} (b-a) [f(a) + f(b)]$$

Example $\int_1^2 e^{-x^2/2} dx$ $f(x) = e^{-x^2/2}$ $a=1$ $b=2$

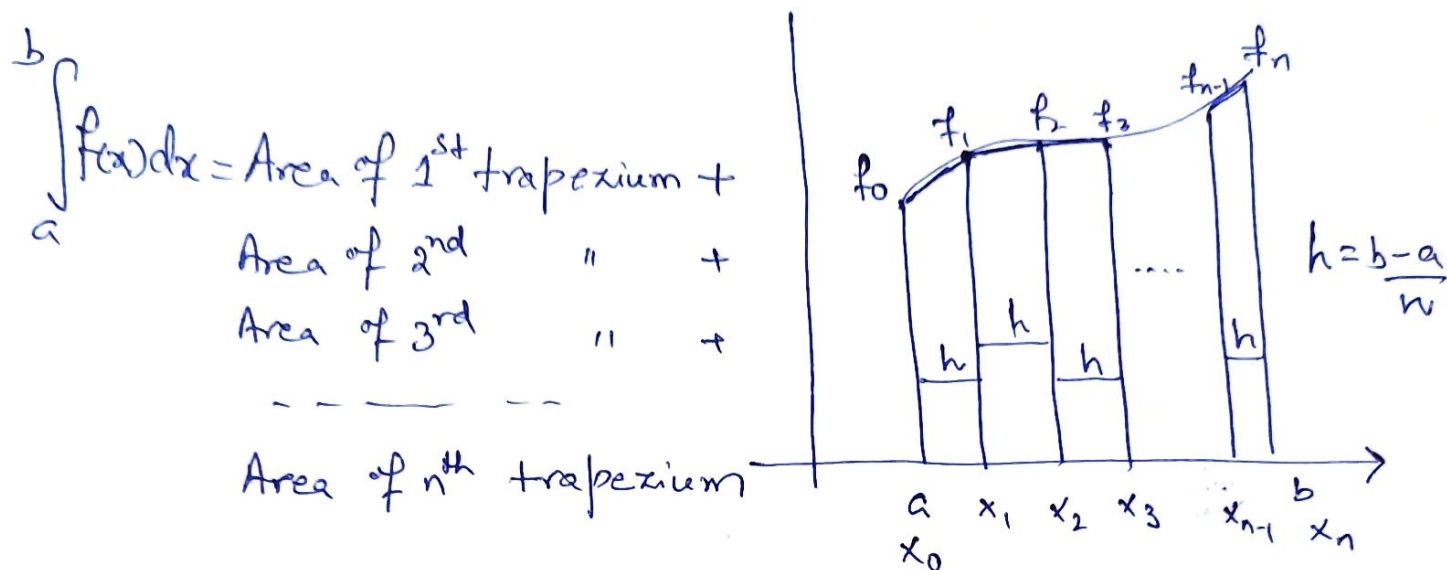
$$\int_1^2 e^{-x^2/2} dx = \frac{1}{2} (b-a) [f(a) + f(b)] = \frac{1}{2} (2-1) [e^{-1/2} - e^{-2}] = 0.37093$$

Ans.

$$\int_0^2 \cosh x dx = \frac{1}{2} (b-a) [f(a) + f(b)] = \frac{1}{2} (2-0) [\cosh(0) - \cosh(2)]$$
$$= 4.76220$$

Divide the area into 'n' numbers of trapeziums.

(8)



$$\int_a^b f(x) = \frac{1}{2} [h] [f_0 + f_1] + \frac{1}{2} h [f_1 + f_2] + \frac{1}{2} h [f_2 + f_3] + \text{--- --}$$

$$\frac{1}{2} h [f_{n-1} + f_n]$$

$$= \frac{h}{2} \left[f_0 + f_n + 2 (f_1 + f_2 + f_3 \text{ --- } f_{n-1}) \right]$$

Formula.

Example

(9)

Approximate $\int_0^2 \cosh x \, dx$ using 4 Subintervals in the trapezoidal rule.

Solution

$$f(x) = \cosh x \quad x_0 = 0 \quad x_n = 2 \quad n = 4$$

$$h = \frac{b-a}{n} = \frac{x_n - x_0}{n} = \frac{2-0}{4} = 0.5$$

<u>x</u>	<u>f(x)</u>
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$x_0 = 0$	$f_0 = 1$
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$x_1 = 0.5$	$f_1 = 1.127626$
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$x_2 = 1.0$	$f_2 = 1.543081$
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$x_3 = 1.5$	$f_3 = 2.352410$
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$x_4 = 2.0$	$f_4 = 3.762196$
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$$\int_a^b f(x) \, dx = \int_0^2 \cosh(x) \, dx$$

$$= \frac{h}{2} [f_0 + f_4 + 2(f_1 + f_2 + f_3)]$$

$$= \frac{0.5}{2} [1 + 3.762196 + 2(1.127626 + 1.543081 + 2.352410)]$$

$$= 3.452107$$

Ans.

Example

$$\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$h = \frac{1}{3}$$

(10)

Solution

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$a = x_0 = 0$$

$$b = x_n = 1$$

$$h = \frac{1}{3}$$

$$h = \frac{b-a}{n} = \frac{x_n - x_0}{n}$$

$$n = \frac{x_n - x_0}{h} = \frac{1-0}{1/3} = 3$$

<u>X</u>	<u>f(x)</u>
$x_0 = 0$	$f_0 = 0.39894$
$x_1 = 1/3$	$f_1 = 0.37738$
$x_2 = 2/3$	$f_2 = 0.31945$
$x_3 = 1$	$f_3 = 0.24197$

$$\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{h}{2} [f_0 + f_3 + 2(f_1 + f_2)]$$

$$= \frac{1/3}{2} [0.39894 + 0.24197 + 2(0.37738 + 0.31945)]$$

$$= 0.33910$$

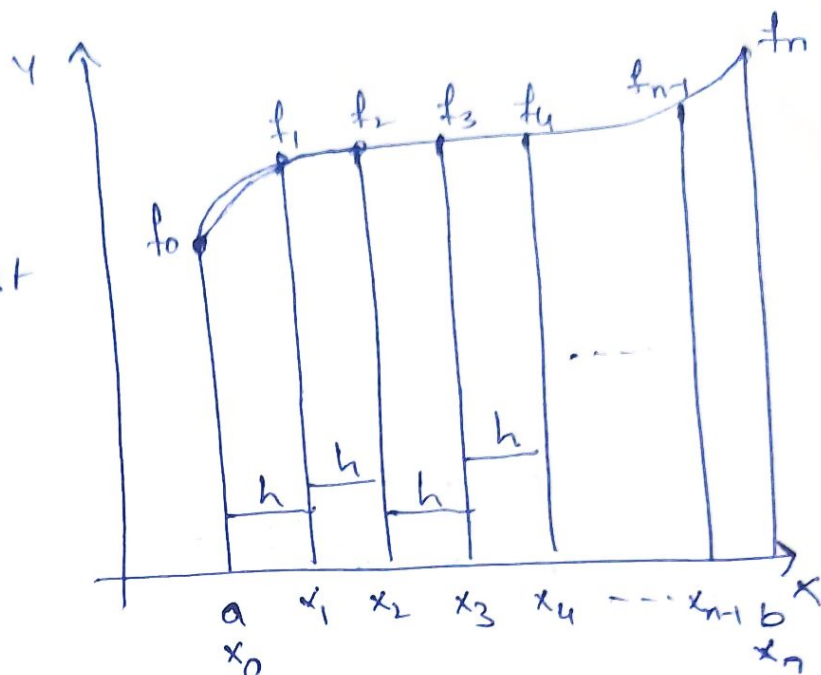
Ans.

Simpson's Rule

Condition

No. of subintervals must be even

$n = \text{even number}$



$$\int_a^b f(x) dx = \int_{x_0}^{x_n} f(x) dx = \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \int_{x_4}^{x_6} f(x) dx + \dots$$

$$+ \dots + \int_{x_{n-2}}^{x_n} f(x) dx \quad \rightarrow \textcircled{1}$$

$$\approx \int_{x_0}^{x_2} P_2^1(x) dx + \int_{x_2}^{x_4} P_2^2(x) dx + \int_{x_4}^{x_6} P_2^3(x) dx + \dots + \int_{x_{n-2}}^{x_n} P_2^{n-2}(x) dx \quad \rightarrow \textcircled{2}$$

where $P_2(x)$ are Lagrange polynomials

Take

$$\int_{x_0}^{x_2} P_2(x) dx$$

(12)

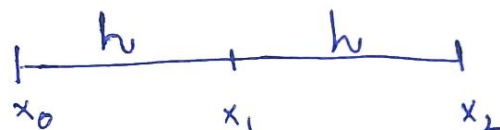
$$\int_{x_0}^{x_2} [L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2] dx$$

$$\int_{x_0}^{x_2} \left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f_2 \right] dx$$

Integration by substitution

(3)

Let $\boxed{x-x_1 = sh}$
 $\boxed{dx = h ds}$



when $x \rightarrow x_0$ $\boxed{s \rightarrow -1}$
 $x \rightarrow x_2$ $\boxed{s \rightarrow 1}$

$$\begin{aligned} x-x_2 &= x-(x_1+h) \\ &= x-x_1-h \\ &= sh-h \end{aligned}$$

$$\boxed{x-x_2 = h(s-1)}$$

$$\begin{aligned} x-x_0 &= x-(x_1-h) \\ &= x-x_1+h \\ &= sh+h \end{aligned}$$

$$\boxed{x-x_0 = h(s+1)}$$

use all in (3)

$$\begin{aligned} & \int_{-1}^1 \frac{(sh)h(s-1)}{(-h)(-2h)} f_0 h ds + \int_{-1}^1 \frac{h(s+1)h(s-1)}{(h)(-h)} f_1 h ds + \int_{-1}^1 \frac{h(s+1)sh}{(2h)(h)} f_2 h ds \\ &= \frac{h f_0}{2} \int_{-1}^1 s(s-1) ds - h f_1 \int_{-1}^1 (s+1)(s-1) ds + \frac{h f_2}{2} \int_{-1}^1 (s+1)s ds \end{aligned}$$

$$\int_{x_0}^{x_2} P_2'(x) dx = \frac{h}{3} [f_0 + 4f_1 + f_2]$$

Now $\int_{x_2}^{x_4} P_2^2(x) dx = \frac{h}{3} [f_2 + 4f_3 + f_4]$

$$\int_{x_4}^{x_6} P_2^3(x) dx = \frac{h}{3} [f_4 + 4f_5 + f_6]$$

⋮

$$\int_{x_{n-1}}^{x_n} P_2(x) dx = \frac{h}{3} [f_{n-2} + 4f_{n-1} + f_n]$$

Therefore from ①

$$\int_a^b f(x) dx = \frac{h}{3} [f_0 + f_n + 4(f_1 + f_3 + f_5 + \dots + f_{n-1}) + 2(f_2 + f_4 + f_6 + \dots + f_{n-2})]$$

formula.

(14)

Example Approximate $\int_0^2 \cosh x \, dx$ using 4 Subintervals in the Simpson's rule.

Solution $f(x) = \cosh x$ $a=0$ $b=2$ $n=4$

<u>X</u>	<u>f</u>
$x_0 = 0$	$f_0 = 1$
$x_1 = 0.5$	$f_1 = 1.127626$
$x_2 = 1.0$	$f_2 = 1.543081$
$x_3 = 1.5$	$f_3 = 2.352410$
$x_4 = 2.0$	$f_4 = 3.762196$

$$h = \frac{b-a}{n} = 0.5$$

$$\int_0^2 \cosh x \, dx = \frac{h}{3} \left[f_0 + f_4 + 4(f_1 + f_3) + 2(f_2) \right]$$

$$= \frac{0.5}{3} \left[1 + 3.762196 + 4(1.127626 + 2.352410) + 2(1.543081) \right]$$

=

Ans.

Example

(15)

$$\int_1^2 \frac{1}{x} dx$$

$$n = 10$$

Solution

$$f(x) = \frac{1}{x}$$

$$a = 1$$

$$b = 2$$

$$n = 10$$

$$h = 0.1$$

<u>x</u>	<u>f(x)</u>
$x_0 = 1$	$f_0 = 1$
$x_1 = 1.1$	$f_1 = 0.9091$
$x_2 = 1.2$	$f_2 = 0.8333$
$x_3 = 1.3$	$f_3 = 0.7692$
$x_4 = 1.4$	$f_4 = 0.7143$
$x_5 = 1.5$	$f_5 = 0.6667$
$x_6 = 1.6$	$f_6 = 0.6250$
$x_7 = 1.7$	$f_7 = 0.5882$
$x_8 = 1.8$	$f_8 = 0.5556$
$x_9 = 1.9$	$f_9 = 0.5263$
$x_{10} = 2.0$	$f_{10} = 0.5$

$$\int_1^2 f(x) dx = \frac{h}{3} \left[f_0 + f_{10} + 4(f_1 + f_3 + f_5 + f_7 + f_9) + 2(f_2 + f_4 + f_6 + f_8) \right]$$

$$= \frac{0.1}{3} \left[1 + 0.5 + 4(0.9091 + 0.7692 + 0.6667 + 0.5882 + 0.5263) + 2(0.8333 + 0.7143 + 0.6250 + 0.5556) \right]$$

$$= \underline{\underline{0.693147}}$$

Ans.